

Scope 1 GHG Inventory — Cement

· FY 2026 · CSI_CEMENT_PROTOCOL_V2 · GWP AR6

Gross Scope 1 direct emissions

24.3 tCO2e

Status: BLOCKED · Data quality: MIXED

Scope 1 breakdown

Clinker calcination	0 tCO2
Bypass dust	0 tCO2
Cement kiln dust (CKD)	0 tCO2
Raw meal TOC	0 tCO2
Conventional kiln fuel	0 tCO2
Alternative fossil kiln fuel	0 tCO2
Non-kiln fossil fuel	0 tCO2
Mobile combustion (owned/controlled)	0 tCO2
Fugitive emissions (refrigerants / SF6)	24.3 tCO2e
Gross Scope 1 total	24.3 tCO2e

Separated buckets (never merged into Scope 1)

Biomass CO2 (memo item)	0 tCO2
Combustion CH4/N2O (non-CSI addendum)	0 tCO2e
Supporting Scope 2 — purchased electricity	0 tCO2
Supporting Scope 3 — bought clinker	0 tCO2
Optional net CO2 (NONE)	n/a tCO2

Combustion CH4 and N2O as CO2e. The CSI Cement Protocol is CO2-only; this line is kept separate from the CSI gross Scope 1 CO2 total and must not be merged into it.

Intensity

Gross CO2 per tonne clinker	n/a kgCO2/t
Gross CO2 per tonne cementitious	n/a kgCO2/t

Factor snapshots (provenance)

CSI default clinker emission factor [CSI Cement CO2 Protocol v2.0]	0.525 tCO2/t_clinker
SF6 (switchgear) GWP (AR6) [IPCC AR5/AR6 100-yr GWP AR5/AR6]	24,300 kgCO2e/kg

Validation

ERROR · missing_facility — A facility must be selected.
ERROR · missing_clinker_production_for_csi_method — CSI clinker-based method requires clinker production. Provide it, or switch to the US EPA cement-based fallback.
ERROR · missing_fuel_quantity — Fugitive entry "Refrigerant / SF6" has no leaked quantity.
WARNING · default_clinker_ef_used — CSI default clinker EF 0.525 tCO2/t used (no plant-specific chemistry).

Calculation trace

Fugitive - Refrigerant / SF6 — 24.3 tCO2

leakedKg x GWP / 1000 · inputs {"leakedKg":1,"gwp":24300,"gwpSet":"AR6","gas":"sf6"}